

Sampling Protocols for New York Fisheries Habitat Management for Wadable Wild Trout Streams

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DRAFT

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1 Introduction

The management of wild trout fisheries is, in practice, the management of the physical and biological components required to support fish reproduction, growth, and survival (Newcomb et al. 2007). Carrying capacity can be positively correlated with habitat suitability (Hulbert 1986, Baldigo et al. 2008, Baldigo et al. 2010, Kratzer and Warren 2013). Correctly identifying limiting factors of habitat suitability associated with wild trout carrying capacity is required to effectively implement habitat interventions needed to achieve the goals of increasing wild trout biomass and abundance within the timeframe of the habitat sub plan (Henson 2026).

This standard operating procedure (SOP) is a technical support document outlining the steps required to monitor success of the Trout Stream Habitat Plan 2026-2036. The first step is a desktop analysis adopted from the United States Forestry Service (USFS) “Stream Inventory Handbook Part 1&2” (U.S. Forest Service, 2012). The desktop analysis is designed to split stream drainages into stream (reach) segments that represent similarities in geomorphic condition. These segments should represent similar fish biomass and abundance across the habitat type.

Within these segments, representative sites will be sampled using the United States Geological Survey (USGS) “Guidelines for Characterizing Fish Habitat in Wadable Streams of the Catskill Mountain Region, New York” (Mulvihill et al, 2003) to characterize habitat variables needed to derive a habitat suitability index (HSI’s) (Raleigh et al., 1982; Raleigh et al., 1984; Raleigh et al., 1986) and estimate the associated fish abundance and biomass. The abundance of natural wood in the stream channel influences a variety of limiting factors of wild trout populations. This SOP also incorporates the Vermont Fish and Wildlife Department (VTFWD) Strategic Wood Addition Handbook protocol (Kratzer, 2020) to quantify wood loading as a limiting factor of wild trout habitat and determine whether supplementation is needed (Kratzer and Warren 2012).

Under this sampling design, habitat and wild trout populations will be measured at the same representative sites. These paired data indicate what factors constrain wild trout carrying capacity and help set priorities for habitat restoration actions. They also provide a baseline against which to measure outcomes for the wild trout fishery and restored habitat.

2 Purpose

The purpose of this protocol is to standardize pre-post habitat characterization and fisheries population estimate sampling efforts across New York State. This sampling

protocol outlines methods to characterize the physical and biological parameters associated with wild trout habitat and estimate wild trout abundance. It outlines a data collection procedure to characterize physical and biological habitat, identify limiting factors of habitat suitability, quantify wild trout population indices, inform habitat restoration strategy, and measure project outcomes. Standardizing these collection methods minimizes variability in the data being collected, establishes pre-post habitat characterization and fish population estimates. The pre – post wild trout population estimates outlined in this protocol are intended to inform management and help justify reduction or cancelation of stocking policy where applicable.

Surveys that do not follow this protocol will not be considered a pre (purpose code 36) or post (purpose code 37) habitat enhancement survey and will not meet the criteria needed to be used in the Trout Stream Habitat Plan.

3 Approach

We plan to implement a stepwise study design (Appendix 1, Figure 1) where a desktop analysis establishes preliminary stream (reach) segments (Figure 1). For these purposes, each stream will be divided into reach segments that are at least 0.5 miles or greater and share similarities in major stream morphology characteristics (Figure 1, Box 1). Reach segment boundaries will be field verified and georeferenced. At this scale we assume wild trout habitat, abundance, and biomass are representative of the identified reach segment. When anthropogenic influence has disturbed a site inside of an otherwise functioning reach segment, each segment may be further broken into segment sections (Figure 1, Box 2). Reach segments and segment sections will provide opportunity to assign a restoration reach, and stable reference (control) reach. A reference reach will need to be identified for each project water but will vary depending on the scale and scope of each project.

Habitat suitability and wild trout population indices will be estimated at the reach segment scale and referenced throughout the project. When applicable, habitat suitability and wild trout population indices may be further identified at the reach segment section scale. Habitat restoration projects will be implemented at reach segment scale when feasible. Once habitat restoration projects have been implemented, post habitat characterization and fish population estimates will inform success of goals and objectives.

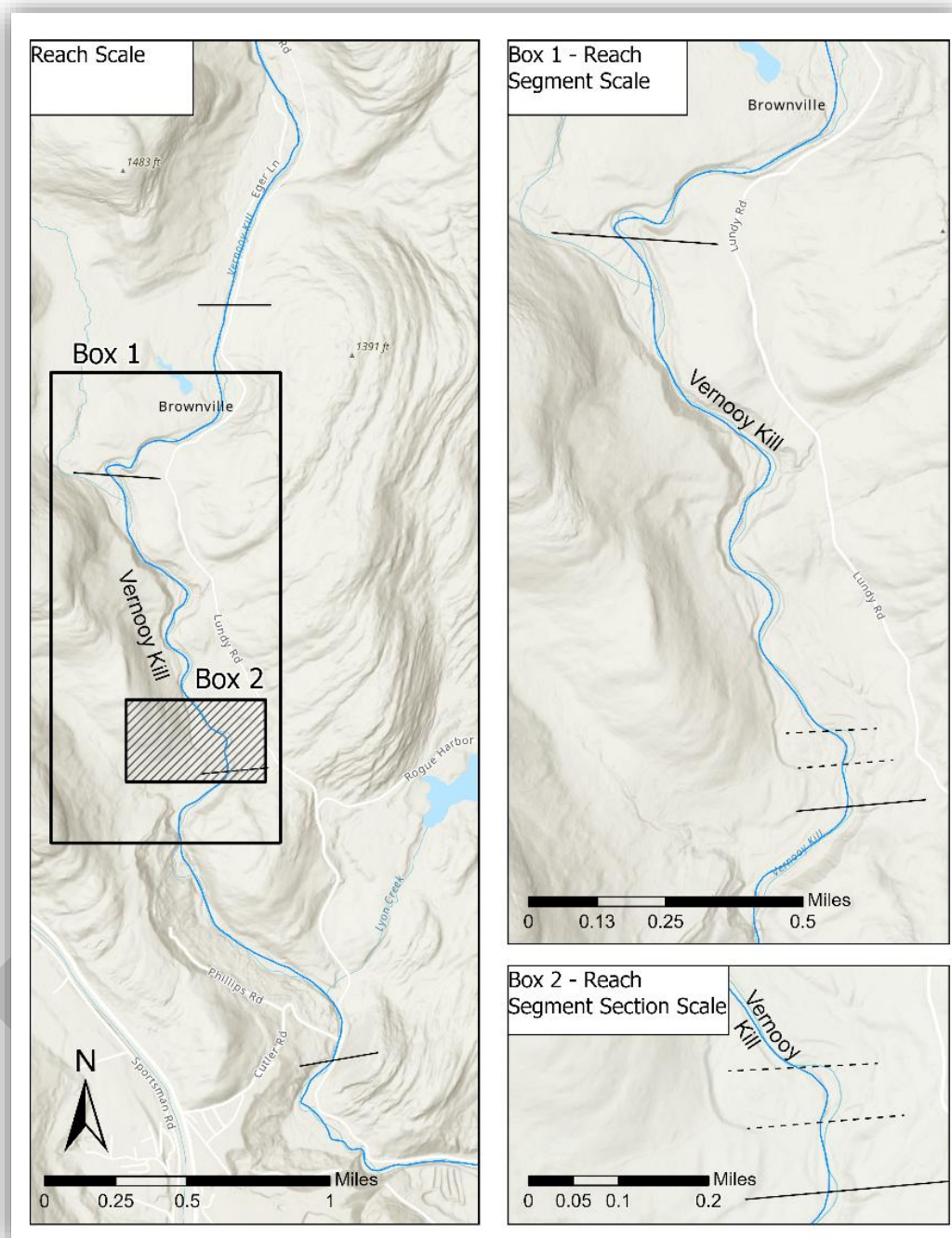


Figure 1: Scale of habitat reach segments (Box 1) and reach segment sections (Box 2) to define representative habitat and fish population estimates pre-post habitat enhancement.

Driven by goals and objectives – The primary management goals defined by the habitat plan will shape the type of data and the scale at which data is collected. The objectives of

each project should identify a limiting factor of habitat suitability and establish wild trout population estimated.

Standardized – A standardized approach to data collection will support a community of practice where outcomes and results can be compared across DEC Regions.

Standardizing data collection will create an opportunity to learn from successes and failures.

Quantitative data collection – The collection protocol design provides a framework for collecting quantitative data that will influence identified project goals and objectives. It will also serve to measure project outcomes of habitat suitability and document wild trout population estimates helping justify the removal of stocking policies.

Statistically valid approach – Collecting quantitative data that is statistically valid provides an opportunity to design habitat projects with the greatest potential to increase wild trout biomass and abundance, justify changes of management category, and termination of stocking policy where applicable.

Repeatable - This protocol provides a standardized quantitative statistically valid method for data collection that minimizes observer bias in characterizing habitat type, level of degradation, fisheries population dynamics, and project outcomes. Collecting data in a standardized repeatable manner promotes the adoption of a before – after – control – impact (BACI) survey design (Underwood 1994) where the fishery and habitat are characterized before and after project implementation.

Step 1: Desktop Analysis.

- Identify scale of the project based on the goals of the habitat sub plan project type.
- Establish study design.

Step 2: In-Stream and Riparian Habitat Evaluation

- Reach Walk
- Field verify desktop analysis
- Georeference of points of interest (POI)

Step 3: Limiting factors survey

- Characterize pre physical and biological wild trout habitat.
- Establish fisheries population indices.

Step 4: Habitat enhancement plan

- Trout stream habitat characterized
- Wild trout population estimates quantified

- Partners and key stakeholders identified
- Project locations identified

Step 5: Project implementation

- Reference individual habitat enhancement management plan for details.

Step 6: Post project monitoring

- Repeat limiting factors survey post project implementation (year 3 & 5).
- Establish post fisheries population indices.

4 Desktop Analysis

Strategies

1. Establish GIS Project
2. Establish preliminary reach segment boundaries.
3. Identify stakeholders
4. Compile supplemental data

Reach segments will be used to direct the limiting factors study, select fish sampling locations, establish a study wide temperature profile, and implement habitat restoration. The desktop analysis is based on standardized methods of the United States Forestry Service (USFS) “Stream Inventory Handbook Part 1” (U.S. Forest Service 2012). All modifications have been made with the intent to best represent the diversity in fluvial geomorphology of New York State and goals of the habitat sub plan.

Most of the desktop analysis will be completed in CO. During the desktop analysis, a GIS project will be created for the use of Central Office (CO) and regional staff throughout the project. The GIS project will be used as a tool for each project water to document survey sites, project locations, deployed temperature loggers, and potential sources of habitat degradation. Once the GIS tool is complete, CO and the region will coordinate field verification and the in-stream and riparian habitat evaluation.

Requirements

- Access to ArcPRO or ArcOnline
- Feature layer files used to derive this information (FIN, Public Access, Topographic layers etc.)

4.1 Establish preliminary reach segment boundaries.

Identify and categorize preliminary habitat stream (reach) segments inside of drainage basin and managed fisheries reach using USGS topographical maps and aerial imagery.

Streams will be segmented based on relative similarities of geomorphic condition. To stratify the managed reach into preliminary reach segments; identify similarities in stream slope, sinuosity, and valley width to represent potential changes in stream type and habitat condition. A geoprocessing workflow has been created with mapped channel gradient, mapped sinuosity, valley width, vegetation types of riparian areas, tributary confluence, and watershed characteristics at a 1:24,000 scale to help delineate preliminary reach boundaries. Known dams, road stream crossings, and land use have been added to help further understand anthropogenic influences on preliminary watershed characteristics. Public access has been added to account for angler access, reach boundaries, and access for sampling.

4.1.1 Preliminary Reach Segments

Using a combination of GIS aerial imagery, and topographic maps with a 20-foot contour interval, at a 1:24,000 scale, select upper and lower reach segment boundaries by identifying break changes in stream slope, sinuosity, and valley width. Reach segments are recommended to be a minimum of 0.5 miles at this scale. Reach sections will be assigned a unique ID. Segments should correspond with large scale geomorphic conditions and Rosgen stream type. Transitional areas between major geomorphic conditions should not be considered part of the reach section when measuring parameters.

Preliminary reach boundaries will be verified during in-stream and riparian habitat evaluation (5.1). During the in-stream and riparian habitat evaluation reach segment boundaries will be accepted or adjusted to accurately represent geomorphic conditions observed in the field.

- If a reach segment (>0.5 miles) is largely impacted by an impoundment, succession of multiple road stream crossings impacting normal geomorphic condition; **or** comprised by all privately owned lands with no agreeable access, it may be decided to assign this area as its own reach segment regardless of geomorphic characteristics. Characterization may be skipped **or** completed depending on the circumstances and the potential effects that they may pose to the fishery. If the need to remove a structure or series of structures has been identified, the influence the structure has on geomorphic condition and habitat will define the scale of sampling. *Examples: impoundments, old degraded habitat structures, and road stream crossing. Determination may be made during the habitat evaluation.

- In the case that stream slope, sinuosity, and valley width are not good indicators of splitting a reach into representative segments of geomorphic condition, reach segments may be adjusted while planning the in-stream and riparian habitat evaluation (e.g. long flat river valleys with limited changes in slope).

4.1.2 Prioritize Key Tributaries

To better understand and prioritize work within the stream drainage, identify tributaries that have the potential to provide either beneficial or detrimental impacts to the managed wild trout fishery.

There is no one size fits all for prioritizing tributaries. As such, presence of indicator species representing water quality derived from the SWFDB (release #103), land use (USGS 2024), NYS dams' layer (NYS DEC Dam Safety), road stream crossings (NAACC), and state lands (NYSDEC) have been added to each GIS project. This step will also heavily rely on regional understanding of the stream drainage.

Tributaries may be treated as additional reach segments and sampled for limiting factors and fish population indices when they are believed to provide biological relevance to the wild population. *Wild trout abundance and/or biomass in tributaries should not be used to justify Wild-Quality or Wild-Premier thresholds of the managed fisheries reach. However, if a tributary meets access, stream size requirements, and abundance and/or biomass thresholds, the decision may be made to change a managed tributary reach to the appropriate management category.

4.1.3 Assign Reach Segments a Unique Hierarchical Identification Number

Starting with the Fisheries Identification Number (FIN) then assigning a unique reach segment identification number (RSID) following flow lines from the top managed fisheries reach to the farthest downstream segment of the project water. Tributaries of interest identified during the desktop analysis or stream walk will have a unique RSID assigned.

*Example: FIN"H-139-14-35", Reach Segment two "RSID-02" = H-139-14-35-RSID-02

4.1.4 Calculate Slope

Once all reach segments have been delineated and assigned an RSID, measure mapped valley length between the upper and lower reach boundaries to the nearest foot. Obtain elevation from the location of each stream segment boundary. Calculate the change in elevation and the difference in slope between the upstream boundary and the downstream boundary (Appendix 2, Table 2).

4.1.5 Calculate Sinuosity

Using the FIN layer flow lines, measure river miles between the upper and lower reach segment boundaries to calculate “mapped reach segment length”. Use “reach length” measured to calculate slope (Appendix 2, Table 2).

4.2 Identify Stakeholders

The stakeholders outreach portion of this step will rely heavily on the relationships between regional fisheries, potential partners, and landowners. Likewise, regional knowledge of the resource will play a vital role in this step of the protocol.

4.2.1 Identify and Collect

Identify any state lands (State Forests, Forest Preserves, and Wildlife Management Units), public access easements, public lands (Town, County etc.) and private landowners to enquire about permissible activities and establish permissions.

All Public Fishing Rights (PFR) Easements and determine what habitat activities may be permissible by each easement. PFR landowners will be contacted once habitat deficiencies are documented and a targeted plan of restoration has been identified.

Forest Management Plans (FMP's) to determine permissible activities within each state forest located within the project drainage area. Contact regional foresters about interest in working with fisheries on habitat restoration.

4.2.2 Key Landowners

Tax parcels will be added to each project water map along the stream banks associated with the stream and its major tributaries. Using these tax parcels, identify landowners within PFR and landowners where no public access is present. Once identified, develop a plan to contact landowners in areas where work may occur.

4.2.3 Identify Partners

Some partners have been identified based on existing projects before the completion of the TSHP. Additional partners who may play a role in project implementation should be identified and documented in each project water enhancement plan.

4.3 Supplemental Data

4.3.1 [USGS Stream Stats](#)

Flow data will be obtained by delineating the drainage area from the farthest downstream section boundary of the managed fisheries reach (U.S. Geological Survey, 2019).

Identify the drainage basin by selecting the farthest downstream reach boundary point using the delineation tool. Download the drainage basin shape file to be used later in GIS, “Peak-Flow Statistics”, “Flow-Duration Statistics”, “Bankfull Statistics”, and “Maximum Probable Flood Statistics. Select all available data to run in the delineation tool.

4.3.2 [North Atlantic Aquatic Connectivity Collaborative \(NAACC\)](#)

NAACC road stream crossings will be exported and georeferenced to be included in the GIS project (Abbot and Jackson 2019). Identify scale needed to include all road stream crossings found within the drainage that may affect the wild trout fishery.

4.3.3 NYSDEC

Threatened and Endangered (T&E) species layers will be added to the GIS project in collaboration with the Rare Fish Unit. DEC State Forest and forest preserve will be added to inform which Forest Management Plans (FMP) may be applicable and inform public access. Public Fishing Rights (PFR) easements will be added to the GIS project.

5 In-stream and Riparian Habitat Evaluation

Objectives:

1. Complete a field verification of preliminary reach segments
2. Assign representative survey locations
3. Catalogue a georeferenced inventory of habitat features
4. Develop a habitat evaluation plan with a summary of the results

The in-stream and riparian habitat evaluation is a reach walk designed to field-verify preliminary reach segment boundaries, establish survey locations, and create a georeferenced catalogue of points of interest (POI). POIs include, but are not limited to previous habitat projects, potential barriers to aquatic connectivity, and visible sources of ground water. While walking the stream, staff will field verify preliminary reach segment boundaries by identifying similarities in meso-habitat (riffle, run, pool, and glides) within each reach segment. Staff will also document all previous restoration projects that will be catalogued in a historic habitat project database (to be created). Field verified reach segment boundaries, POI’s, survey locations, and preliminary habitat conditions will be reported on in individual habitat enhancement plans.

When anthropogenic influences cause the geomorphology to display characteristics that are **not** representative of the greater reach segment, reach segments may be further broken into reach segment sections based on similarities of meso-habitat or observation of homogeneous habitat, representing similar geomorphic condition. Examples include,

but are not limited to, excessive erosion from poor land use practice, road stream crossings or dams presenting a barrier to aquatic passage. Stream segment sections with a high proportion of degradation from anthropogenic influence will be identified and documented as potential restoration reaches. Additionally, stream segment sections that may serve as undisturbed control reaches and/or stable reference reaches will be identified and documented. The reach walk is based on standardized methods of the United States Forestry Service (USFS) “Stream Inventory Handbook Part 2” (USFS Pacific Northwest Region 2012). All modifications have been made with the intent to best represent the diversity in geology of New York State.

Equipment Required

- Smartphone or tablet with camera and *Field Maps* application loaded (must be linked with GIS habitat project).
- 300’ measuring tape
- 100’ measuring tape
- Stadia Rod
- Write in the rain notebook

5.1 Field Verification

Field-verify preliminary reach segment boundaries by visually confirming if the georeferenced boundaries accurately represent a noticeable change in geomorphologic conditions. All georeferenced reach boundaries will be documented on *Field Maps*. Beginning and ending reach segment boundaries should also be recorded by hand on a separate data sheet in UTM format.

5.1.1 Field Verify Preliminary Reach Boundaries

Preliminary reach segment boundary point accurately represents the location of change in geomorphologic condition. If the preliminary reach segment boundary point accurately represents the location of change in geomorphologic condition, acquire a georeferenced GIS data point, and select option “Accept Boundary”. If the preliminary reach segment boundary point does not accurately represent the location of fluvial geomorphologic change, select option “Reject Boundary” and acquire an alternative georeferenced GIS data point.

If the reach segment boundary has been rejected, visually identify the location that accurately represents a change in geomorphic condition and assign a new georeferenced reach boundary GIS data point. * This location should be close by, within a few hundred

yards upstream or downstream at the slope break. Record UTM coordinates on the datasheet before leaving location.

If the preliminary reach segment boundary point does not accurately represent the location of change in geomorphologic condition, **and** there is no observed change between preliminary reach segments, select option “Reject Boundary” and add a descriptive note to document **why** this option was selected.

5.1.2 Characterize

Assign each segment or segment section a Rosgen stream type and take notes documenting the type of mesohabitat present. Mesohabitat and Rosgen stream type will be used to categorize segment sections by either assumed “functioning” at ecological capacity, or “degraded” ecological capacity.

While identifying segment or segment sections, take descriptive notes to further categorize mesohabitats that will be used when assigning BACI study locations.

*Familiarity with Rosgen stream classification will help further infer “functioning” or “degraded” functional capacity based on assigning stream type. * Visually estimate based on a combination of presence absence of mesohabitat, channel shape, and dominant substrate type.

5.2 Visually Identify Changes in Geomorphic Condition

A visual observation of change in stream slope, and valley width or bankfull channel width should be identifiable at the preliminary reach segment boundaries at this scale. The change in slope and valley width should be observable near this location. If no change in geomorphic condition is observed, deny the reach boundary and make a note that no change of geomorphic condition was observed. *Change in stream slope should not be verified with the use of a clinometer, digital slope range finder, or measurements in valley width or bankfull channel width during this step.

5.2.1 Document Anthropogenic Influences

Identify potential changes in geomorphic condition or mesohabitat caused by anthropogenic influences. Reach segment sections that share similarities in physical habitat represented by changes in flow pattern, channel shape, and scour pattern (pool to pool spacing). Identifying and documenting reach segment sections will help identify potential locations for habitat restoration. * Road stream crossings identified as important should be assessed separately using the NAACC survey protocol.

5.3 Assign Representative Survey Locations

Survey sites must represent the habitat observed in the reach segment. Therefore, while there is no predefined length for survey sites, they must include a full sequence of pool-riffle-pool-riffle. Sites will be used in the limiting factors survey and sampled pre-post habitat project. In the case that a reach section is best represented by a long flat featureless channel, survey sites should be represented by ≥ 20 mean channel width or a maximum length (Appendix 2, Table 3). Site selection should always be made so that measurements accurately represent the reach segment or reach segment section depending on representative scale. Site boundaries should not extend into non representative habitat to meet the minimum required site length. As such, they must not include multiple reach segments.

5.3.1 Locate Visually Representative Sites

Visually identify a length of stream that is visually representative of the greater stream segment. To accomplish this, walk the reach segment or segment section and observe changes in habitat, fluvial geomorphology, and tributary influence. Survey locations should not include beaver dams, log jams, natural barriers, or influence from anthropogenic activities. *Exception: If the goal of the survey is to characterize the influences on the fishery by beaver dams, log jams, barrier, or anthropogenic influence is part of the dominant stream type **OR** the site has been selected as part of a reach segment section.

5.3.2 Flag Sampling Site Boundaries

Mark survey site with semi-permanent markers at the upper and lower boundaries of the survey site. Obtain NYTMN/NYTME of the survey boundaries. The boundary should begin at the downstream end of the starting stream feature. Assign survey site numbers once sampling sites are verified.

5.4 Catalogue a Georeferenced Inventory of Habitat Features

Georeference Points of Interest (POI) on the *Field Maps* app, using the map titled “Statewide habitat points”. Select the collection category that accurately represents the POI being georeferenced. Assign each habitat feature a georeferenced point with the appropriate title and collect associated metadata.

5.4.1 Characterize Habitat Features

All points should be characterized and assigned a descriptive title so that it may be referenced in the future. Assign each feature a descriptive category that best represents the type of data point being collected. *Example: historic habitat project, potential barrier

to aquatic connectivity, ground water source, bank degradation, channel armoring, trash pick-up, temperature reading etc. Be descriptive, add relevant metadata to the “notes” segment. *Example: “Historical CCC project no longer functioning and exposed rebar presenting potential safety risk to anglers”.

5.4.2 Photograph Habitat Features

All feature points will have one descriptive photo that captures as much of the project as possible attached before submitting. Pictures will be used document condition and should encompass the entire project in one or two concise pictures for the georeferenced feature point. * Take many additional pictures separately to track progress and document pre-post site conditions.

6 Limiting Factors Survey

Objectives:

- Characterize wild trout habitat suitability
- Quantify fish population indices

The limiting factors survey is a paired habitat characterization and wild trout population estimate. The variables associated with the limiting factors study will characterize physical and biological habitat representing a level of suitability (Appendix 1, Figure 2) that is positively correlated with carrying capacity representing a wild trout’s ability to carry out its life history (Raleigh, 1982; Raleigh and others, 1984; Raleigh and others, 1986).

The associated variables referenced in this sampling protocol are grouped into three potential main sampling categories associated with the scale at which they are biologically relevant under this study design (Appendix 1, Table 1). The sampling categories drive the scale at which measurements of biological and physical habitat variables will be collected: variables that are general to each reach, variables specific to each reach segment and variables that may be measured at the reach segment section when applicable. The collection methods referenced herein are based on methods and protocols developed and implemented by USGS in New York trout streams (Baldigo et al. 2010; Ernst et al. 2010; Mulvihill et al 2003).

Establishing pre and post habitat conditions and wild trout biomass estimates will set baselines and document project outcomes. Survey locations will function as either a treatment or control within reach segments or segment sections. All surveys should be completed during base flow conditions. Surveys should be conducted away from the

influence of a bridge or culvert (>100m) unless the road stream crossing is part of the project. * Refer to reach segments and segment sections established in the in-stream and riparian habitat evaluation portion of this plan (5.2). Sampling habitat and fish are outlined in reverse order herein. Fish sampling is intended to take place before collection of habitat variables to ensure that fish are not pushed out of the sample site.

Equipment Required

- Smartphone or tablet with camera and *Field Maps* application loaded (must be linked with GIS habitat project).
- Range finder
- Clinometer
- 300' measuring tape
- 100' measuring tape
- Stadia Rod
- Write in the rain notebook
- Gravelometer
- Temperature loggers
- Flow prob
- Modified convex spherical densiometer
- Electrofishing Backpack (s)
- Scap nets
- Block nets
- Fish holding bins
- Measuring board
- Fing board
- Digital scale
- Bubble level
- Surveying pins
- Digital thermometer
- YSI (Calibrated annually)
- HACH Kit
- Temperature data loggers (HOBO)
- Rebar
- Sledgehammer
- Slotted PVC
- ZIP Ties

6.1 Characterize Wild Trout Habitat Suitability

Reach level variables will be taken quarterly to represent water quality variables associated with different times of the year. Reach segment, and segment section variables will be collected at the paired habitat suitability characterization and fisheries population estimate survey sites one- or two-years pre project implementation. Post project monitoring will consist of repeating a paired habitat suitability characterization and fisheries population estimate of reach segment and segment section variable three- and five-years post project completion.

Reach segment and segment section variable characterization should be completed during summer base flow conditions. Habitat suitability measurements at the reach segment section scale are obtained by completing 11 transects throughout the survey site. Survey sites should include two sequences of mesohabitat (pool, riffle, pool riffle). Variables associated with reach level scale will be measured outside of the survey site. Space between transects should be obtained by measuring site length and divided by the 11 transects to inform the distance between transects and limit observer bias.

6.1.1 Water Quality

6.1.1.1 Mean Daily / Maximum Daily Water Temperature During Low Summer Flows & During Fish Embryo Development Period.

Temperature loggers will be placed at the upper and lower reach segment boundaries and in the upper and lower segments of key tributaries. Loggers will be set to record a temperature every half hour. Loggers will be deployed for the entirety of the study, checked quarterly, and remain in the streams for 3 – 5 years post project completion. Water temperature measurements should be collected quarterly from the temperature loggers.

6.1.1.2 Annual Minimum Water pH

A water sample at the downstream most reach segment boundary will be taken at quarterly check and brought back to the regional DEC office where a YSI and or HACH kit is used to measure pH.

6.1.1.3 Dissolved Oxygen (DO)

DO measurements should be obtained at each temperature logger location and will be used to estimate temperature DO relationships during quarterly checks. Measure DO and temperature with a YSI and write down parameters at each temp logger site.

6.1.2 Wood Loading Levels

6.1.2.1 Percent In-stream Cover During Summer Low Flows.

Within each defined reach segment, three locations should be randomly selected far enough into the reach segment to ensure the influence of the transitional areas of major stream morphology (reach segment boundaries) are not captured and influencing the characterization of each reach segment. At each location, measure a 100' length of stream. Within this length of stream, measure slope, bankfull width, and note the first and second most dominant substrate type. Count the total number of pieces of large wood ($\geq 4"$ in diameter and $\geq 3'$ in length) within the 100' length of stream and bankfull channel width.

Wood loading levels may be measured during the reach walk from step 5 or the limiting factors survey depending on staff time and availability.

6.1.3 Geomorphic Condition

Habitat characterization should be completed during base flow conditions and should be completed on the same day (following) the fisheries population estimate survey is conducted. Characterizing habitat will be further broken into categories, pool class and site rating, stream discharge, and geomorphology (Mulvihill et al, 2003, Raleigh et al., 1982, Raleigh et al., 1984, Raleigh et al., 1986).

6.1.3.1 Pool Class & Site Rating

Measure each pool and assign pool class (Appendix 2, Table 4). Each pool rating is associated with depth, flow and cover present. Pool classes described in detail and have been adopted from the HSI (Raleigh et al. 1982). Measure and record the total length of each pool in the survey site. Assign site rating depending on the proportion of pool classes to the survey site (Appendix 2, Table 5).

6.1.4 Transects

Geomorphologic habitat suitability measurements are obtained by completing 11 equally spaced transects throughout the survey site (Appendix 2, Figure 2). Site length measurements should be taken down the right stream channel (looking downstream) from the top site boundary to the bottom site boundary. Transect locations should be assigned by dividing site length by the 11 (transects) to establish transect locations and limit observer bias. Measure and mark transect location with flagging labeling ascending from transect 1 (T1) upstream, to transect 11 (T11) downstream direction within the site boundaries. The following steps will be repeated at each transect location

6.1.4.1 Position Transect Tape.

At transect station, extend a measuring tape perpendicular to flow at right back and left bank transect flags. Pin measuring tape to right and left bank at mean channel width (MCW) measuring the active channel at high flow conditions.

6.1.4.2 Record habitat type.

Record the meso-habitat type (pool or riffle) intercepted most by the transect tape.

6.1.4.3 Establish sampling locations.

Identify and measure mean wetted channel width (WCW) by subtracting the total distance between the edge of water on the right bank and the left bank from MCW (Appendix 1, Figure 3). Divide the WCW by 8 to derive the sampling point intervals. P1 is located at the conjunction of the right bank and right water's edge, P9 is located at the left bank and left water's edge, and P2 through P8 are spaced at equal intervals along the tape between these two points. Sampling point intervals are rounded to the nearest 0.1ft.

If channel features (boulders, point bars, islands or other obstructions) are present above the water's surface, account for them by measuring the distance protruding above the water surface (e.g. 3ft boulder between P3 and P5). Once measured, remove the value measured of protruding channel feature from the right bank water's edge starting point (P1) interval so that it accurately reflects wetted width.

6.1.4.4 Measure temperature, water depth, and velocity.

Measure water temperature with a field meter stick on the streambed at all sampling intervals along each transect. Use a stadia rod to measure depth at each sampling interval along each transect. Measure velocity based on the number of revolutions a flow probe makes at each sampling interval point. If the water is too shallow or obstructed by a channel feature, note velocity as TS. If it is too weedy to obtain a measurement, note velocity as TW.

6.1.4.5 Streambed Substrate.

Measure dominant substrate size using the Wolman Pebble Count method described herein (Wolman 1954, Bevenger and King 1995). A Wolman pebble count consists of randomly selecting substrate along multiple transect within a defined site. Ten randomly selected pieces of substrate will be collected along all 11 transects.

Start by selecting a piece of substrate by randomly selecting the first piece that your finger touches when bending over and looking forward. **DO NOT** look down while selecting substrate, this introduces observer bias to the survey. Measure each particle of substrate with a gravelometer and record the size of each particle (Appendix 2, Table 6). Randomly select substrate 10 times along the active channel while keeping standardized spacing closely mirroring distance between sampling intervals.

Estimate embeddedness to the nearest 10 percent interval by visually estimating the proportion of large gravel, cobble, and boulders surrounded by smaller gravel or silt and sand.

6.1.4.6 Canopy Cover Characteristics

Measure the proportion of shade on the stream between 1000 and 1400 hours with a modified convex spherical densiometer. The convex spherical densiometer should be modified with tape to form a “V-shape” so that 17 of the possible 37-line intersections are visible (Appendix 1, Figure 4). Four canopy measurements should be taken at each transect location. Densiometer measurements should be taken roughly one foot above the water’s surface (Burres 2010). Four measurements should be taken at three locations along each transect. Measurements should be taken at right water’s edge, left water’s edge, and center of each transect location facing upstream, facing downstream, facing right bank, facing left bank. Use the same relative position for all four measurements to encompass vegetation cover. Count the number of possible point intersections with vegetation cover out of 17 for each measurement. The proportion of tree cover and low shrub cover should be captured in the reflection of intersect points on the special densiometer.

Canopy cover will be converted into a percentage by dividing total possible canopy point intersections (204 per transect) measurements taken with the total number of canopy intersects measured (2,244 total point intersects) with across all transects multiply by 100 (Appendix 2, Table 2).

6.1.4.7 Characterize Bank and Riparian Condition

Visually estimate bank characteristics of both the right and left bank along a 2ft wide corridor that starts where the transect meets the water’s edge and extends shoreward to the highest point of annual flow conditions (find wrack line, edge of rooted vegetation, or other stream characteristics informing annual high water).

Visually estimate dominant bank substrate type of both stream banks using the same size classes from the Wolman pebble count.

Visually estimate dominant vegetation cover of both stream banks as bare ground, grass, shrubs, or trees.

Visually estimate bank stability of both stream banks as 1 (0-24% stable), 2 (25-49% stable), 3 (50-74% stable), 4 (75-100% stable).

6.1.5 Identify and Characterize Ground Water

While this protocol outlines methods for georeferencing visible ground water sources, additional steps to quantify spring seeps and hyporheic flows may be necessary where temperature is limiting (personal communication Barry Baldigo, 2026).

6.2 Fish Sampling

6.2.1 Time of Sampling

All fish sampling is intended to take place during base flow conditions. Water temperatures should be below <71°F to limit stress causing mortality of wild salmonids (Wehrly 2007). The fish survey portion of this protocol should be completed before the habitat characterization to ensure fish have not been pushed out of the survey site.

6.2.2 Site Selection

Refer to site selection methods on page 16 section 5.3 of this SOP. Site number, location, and coordinates in NYTM format will be recorded on SL data sheets.

6.3.3 Operation and Deployment (Dunham et al. 2009)

Population estimates used to quantify the wild trout population rely on the assumption of sampling a closed system (White et al. 1982). As such, block nets with a lead line and 5-7mm bar mesh are deployed at the upper and lower boundaries of the survey site. All survey sites will be sampled with electrofishing gear in an upstream direction standardized by number of gear deployed and number of scap netters. All surveys should consist of three successive passes (reps) to ensure sufficient confidence in the depletion. All trout captured will be kept aside in totes separated by rep with methods in place to keep fish out of the sun and water oxygenated to limit stress on fish while reps are finished. Seconds shocked should be recorded from each backpack shocker and recorded at the end of each rep.

6.3.4 Effort

Fish sampling effort will be recorded in catch per unit effort (CPUE) by seconds shocked, **AND** in fish captured per area sampled to quantify abundance. Fish sampling may be further standardized by the total amount of backpacks and netters deployed (Dunham et al. 2009) per mean channel width sampled (Appendix 2, Table 7).

6.3.5 Target Species

Wild trout are the primary targeted species to be collected under this SOP. All trout species will be identified and measured by total length in millimeters and weight in grams and will be recorded on an individual fish (IF) form. Estimate the number of non-game fish and record on the bulk fish (BF) form. All wild trout should be categorized as wild (W) in the "Wild" field on the IF. * If stocked fish are observed (clipped fins or other indicators), make sure they are categorized as stocked (S) in the "Wild" field on the IF.

The Rare Fish unit (RFU) should be consulted prior to all surveys to identify potential rare fish that might be in the survey water. If there is interest in collaboration with the Rare Fish Unit, voucher collection protocol should be followed if a fish is captured and not easily identified. If there are no RFU priority fish, bulk fish may be estimated based on observations made from the first rep of all non-trout species. If there are RFU priority fish

present, discuss with the RFU the possibility of IDing all fish captured or looking out for individual rare fish while surveying is being conducted.

7 Habitat Evaluation Plan

Each project water will have a habitat evaluation plan written after the desktop analysis, and in-stream and riparian habitat evaluation are completed. Habitat evaluation plans will be added to throughout habitat work and post monitoring. Each habitat evaluation plan will serve to document all work completed through the project and provide future direction.

Each project water will have a pre and post habitat evaluation plan to document conditions and work completed.

7.1 Habitat Characteristics

Each plan will report on initial stream characteristics, survey locations, land use, Rare species, state regulated wetlands, water quality monitoring locations, points of interest, stakeholders, and partners will be documented. Habitat suitability pre restoration results at each reach segment will be reported. Areas where habitat suitability may be addressed with habitat restoration / enhancement and recommendations to address limiting factors will be presented. This portion will be built out farther after the first year of sampling has been completed.

7.1.1 Habitat Suitability Index (HSI)

HSI are calculated for each reach segment and segment sections surveyed (Appendix 1, Figure 5). Each variable associated with trout's life a life history HSI is based on a proportional scale where 1 or 100% is optimal and > 0.4 or 40% is considered a limiting factor.

Identification of limiting factors with Bayesian Hierarchical models or Maximum likelihood estimates.

7.1.1.1 Adult Trout Habitat Variables

Adult trout habitat suitability will be calculated based on measured variables associated with average thalweg depth, % instream cover, % pools, pool class.

7.1.1.2 Juvenile Trout Habitat Variables

Juvenile trout habitat suitability will be calculated based on measured variables associated with % instream cover, % pools, pool class.

7.1.1.3 Fry Stage Trout Habitat Variables

Fry stage trout habitat suitability will be calculated based on measured variables associated with % substrate size, % pools, % riffle fines.

7.1.1.4 Embryo Stage Trout Habitat Variables

Embryo stage trout habitat suitability will be calculated based on measured variables associated with average max temp, average minimum DO, average water velocity, average substrate size, % riffle fines.

7.1.1.5 All Life Stages of Trout Variables

Habitat suitability for all life stages will be calculated based on measured variables associated with average max temperature, average minimum DO, pH, average minimum base flow, dominant substrate type, average % vegetation, % streamside vegetation, % riffle fines, % midday shade

7.2 Data Analysis

Standardized methods for habitat characterization and wild population estimated will be added to this section. All data collected from the desktop analysis, in-stream and riparian habitat evaluation, and limiting factors survey will be compiled as variables of maximum likelihood estimates to inform habitat work directed at addressing limiting factors.

7.2.1 Wild Trout Population Estimates

K-pass collection methods deployed will determine fish population estimates. This will be built in RStudio and added in a future version of this protocol.

7.2.2 Wild Trout Population Indices

Sampling methods will allow us to quantify population indices of biomass (lbs/acre), relative abundance (CPUE fish/mile), relative weight, length frequency, and proportional size distributions. These indices will pair with habitat suitability index's and will be built in RStudio and added to a future version of this protocol.

7.3 Project Monitoring

Project monitoring and updated sampling takes place after each phase of the projects are completed. Project monitoring will be conducted two years and five years post project implementation.

All post surveys will be completed in the same locations with the same amount of effort as the pre surveys. All post surveys will mirror the habitat measurements and three pass

depletion carried out in the pre survey. Post project sampling takes place year 3 and year 5 after project completion.

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9 Appendix 1 Figures

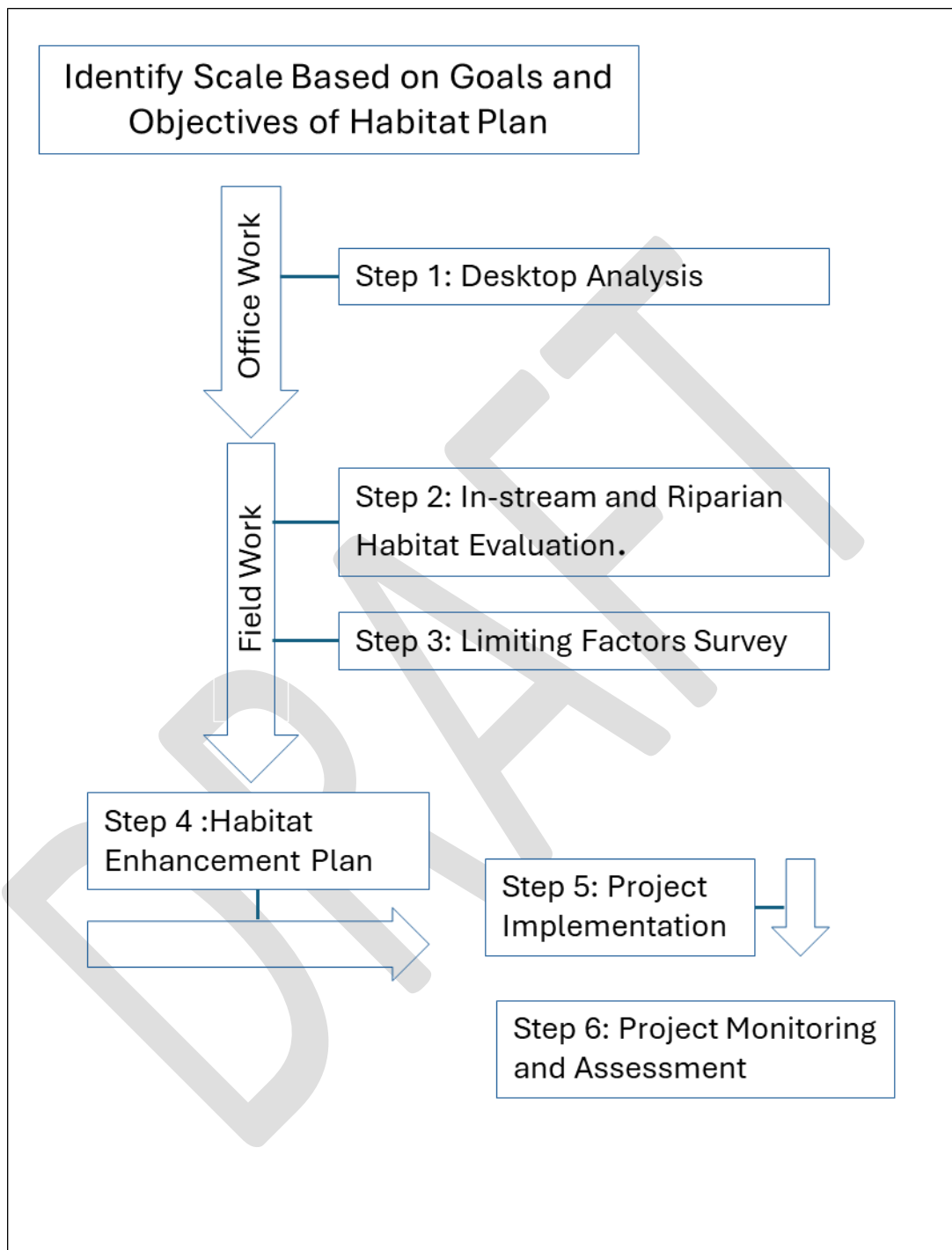


Figure 1. Plan overview of the steps plan outlined in this protocol.

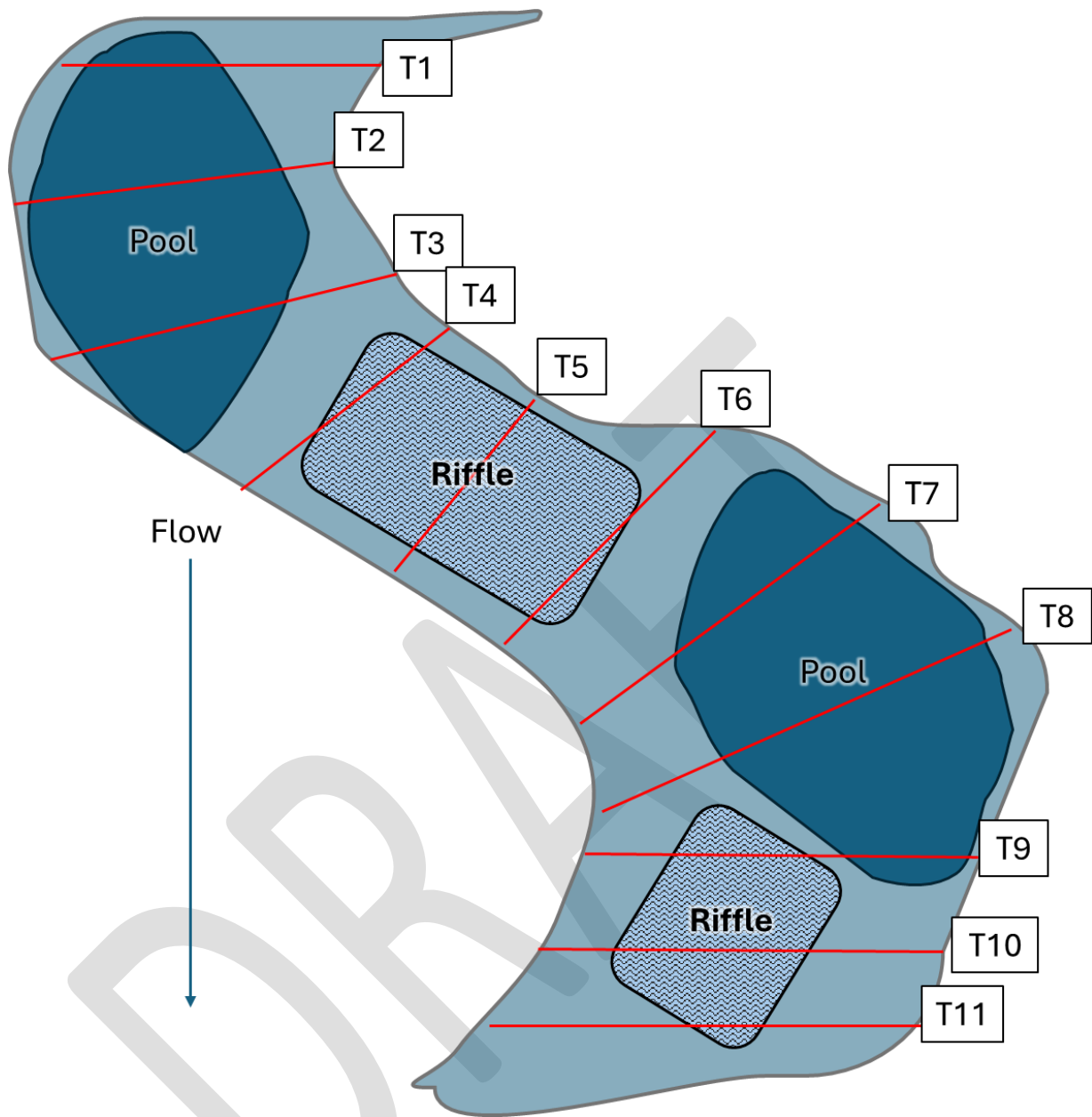


Figure 2. Longitudinal profile of a survey site with 11 equally spaced transects of geomorphologic habitat measurements. Transect locations should be assigned with flagging and labeled in ascending order from transect 1 (T1) to transect 11 (T11) in downstream direction.

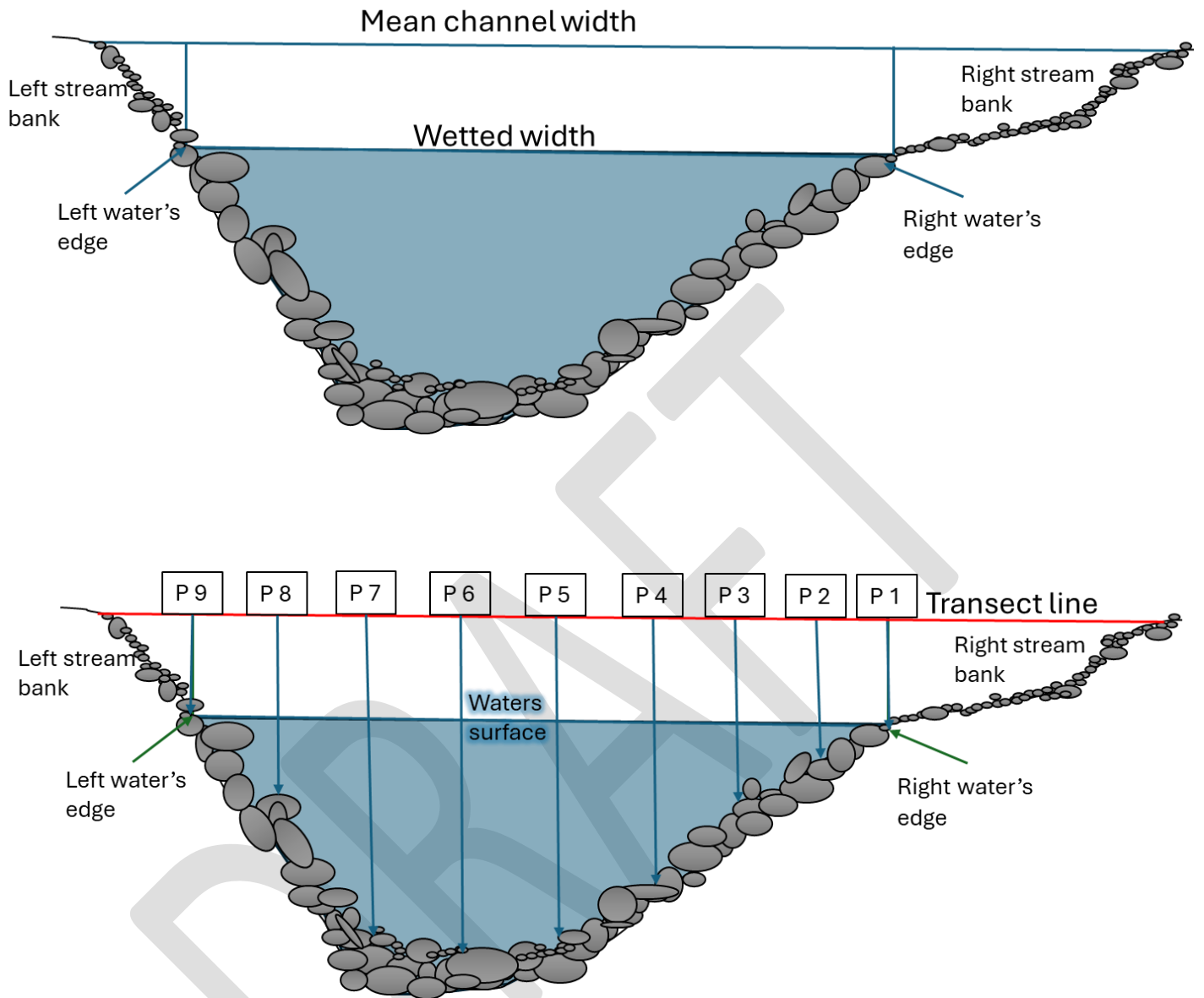


Figure 3. Stream cross section depicting the location of measurements of transect points. P1 is located at the conjunction of the right bank and right water's edge, P9 is located at the left bank and left water's edge, and P2 through P8 are spaced at equal intervals along the tape between these two points.

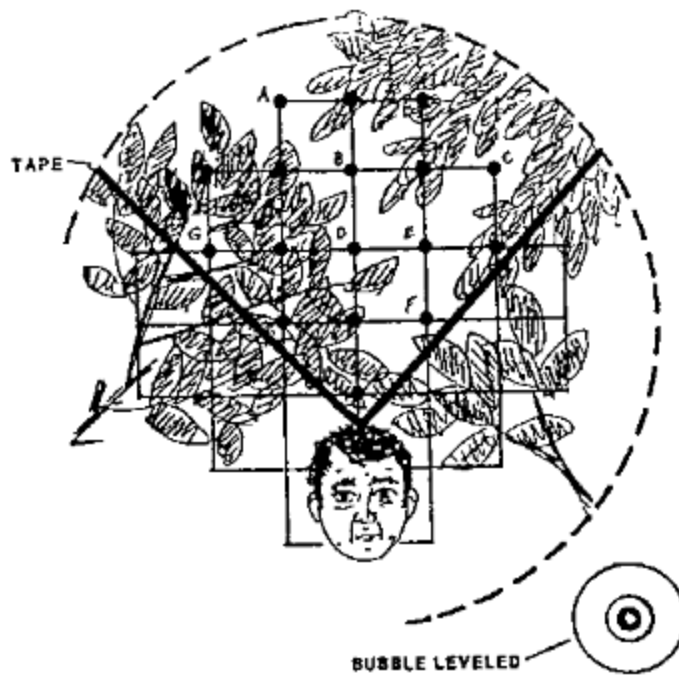


Figure 4. Convex spherical densiometer should be modified with tape to form a “V-shape” so that 17 of the possible 37-line intersections are visible (Mulvey et al. 1992)

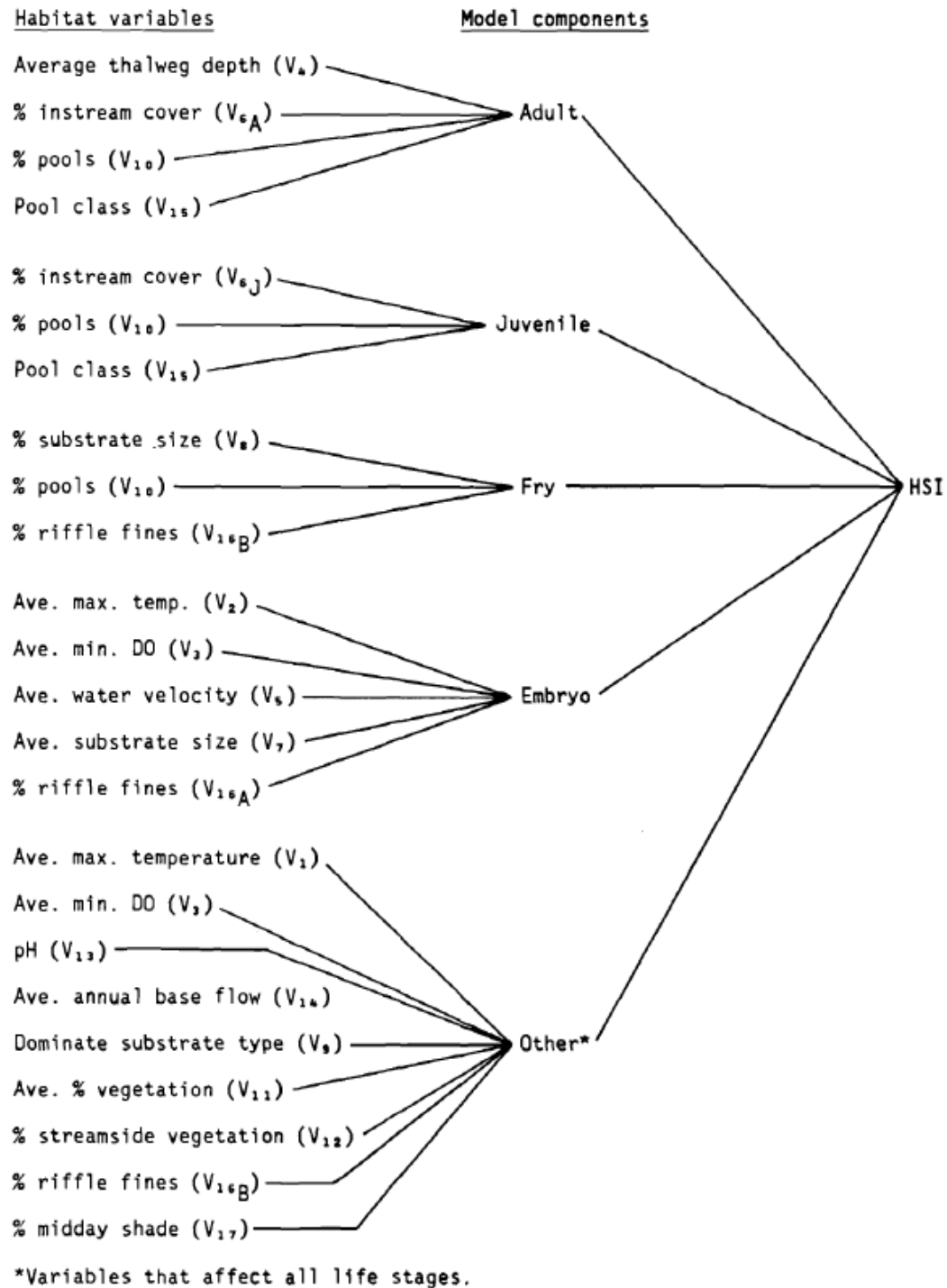


Figure 5. Habitat suitability index based on variables influencing all trout life stages (Raleigh et al., 1982).

10 Appendix 2 Tables

Table 1: Habitat suitability variables and associated scale of measurement.

Scale	Variable	Description
Reach Level Variables	1	Mean daily minimum DO during low summer flows
	2	Annual minimum water pH
Reach Segment Variables	3	Mean daily maximum water temperature during low summer flows
	4	Mean daily maximum water temperature during fish-embryo-development period
	5	Percent in-stream cover during summer low flows (VT SWA)
Reach Segment / Section Variables	6	Wild trout population estimate
	7	Mean water velocity during embryo development period
	8	Percentage of average annual flow represented by mean base flow
	9	Mean thalweg depth at low summer flows
	10	Mean size of riffle substrate material with intermediate diameter from 2 to 80 mm
	11	Percentage of substrate in reach with diameter of 100 to 400 mm
	12	Dominant substrate type in riffles
	13	Percentage of reach occupied by pools at low summer flows
	14	Percentage of bank that is vegetated by grass, shrub, or trees
	15	Percentage of bank with either vegetation or rocky ground cover
	16	Pool class rating during low fall flows
	17	Percentage of riffles and runs represented by fine material (diameter < 0.063 mm)
	18	Percent of stream shaded from 1000 - 1400 hours

Table 2. Formulas used to calculate variables associated with geomorphic condition.

Variable	Formula
Slope (S)	$S \% = \text{Change in Elevation (Rise)} / \text{Valley Length (Run)} * 100$
Sinuosity (k)	$K = \text{mapped reach segment length} / \text{valley length}$
Flow (V)	$V = 0.9604 \times (\text{number of revolutions/time in seconds}) + 0.0312$
Canopy Cover	$\text{Canopy Cover} = \text{Potential convex spherical densiometer intersects measured (n)} / \text{Total canopy intersects (2,244)} * 100$

Table 3. Maximum site length for flat featureless reach segments based on stream width.

Stream Width (ft)	Max Survey length (ft)
<15	250
15 - 30	500
31 - 60	750
>60	1000

Table 4. Pool class descriptions associated with depth and cover for adult trout sp.

Classification	Description
First-class pools	Large enough and deep enough to provide a low-velocity resting area for many adult trout. More than 30 percent of the bottom is obscured by deep water and surface turbulence, and by structures such as logs, debris, boulders, overhanging bank, or vegetation that provide shelter in which fish can rest, hide, or feed. The greatest pool depth exceeds 4 ft in streams less than 13 ft wide and exceeds 6 ft in streams more than 13 ft wide.
Second-class pools	Intermediate size and depth and provide a low-velocity resting area for a few adult trout. From 5 to 30 percent of the bottom is obscured by deep water, surface turbulence, and structures. Large eddies behind boulders and moderately deep areas under banks and vegetation are typical.

Third-class pools Small and shallow and provide a low-velocity resting area for one or two adult trout.

Less than 5 percent of the bottom is obscured by deep water, surface turbulence, or structures.

Wide, shallow stream areas and small eddies behind boulders are typical.

All pools that have no cover at all are considered third class.

Table 5. Site rating descriptions associated with the proportion and class of pools present.

Rating	Designation
A	More than 30 percent of the reach consists of first-class pools
B	More than 10 percent, but less than 30 percent, of the reach consists of first-class pools. OR More than 50 percent of the reach consists of second-class pools
C	Less than 10 percent of reach consists of first-class pools, AND less than 50 percent of the reach consists of second-class pools.

Table 6. Substrate size class based on size associated with a Wolan pebble count.

Size Class	Size Range (mm)
Sand	<2
Very fine gravel	2 - 4
Fine gravel	5 - 8
Medium gravel	9 - 16
Coarse gravel	17 - 32
Very coarse gravel	33 - 64
Small cobble	65 - 90
Medium cobble	91 - 138
Large cobble	129 - 256
Small boulder	257 - 512
Medium boulder	513 - 1024
Large boulder	> 1025

Table 7. Effort for sampling adult wild trout by gear and netters deployed based on the size of the survey water. Waters >60' may require multiple sampling types.

Stream Width (m)	Gear	Wand / Backpacks	Scap Netters
<15	Backpack	1 - 2	2-3
15 - 30		2 - 3	3-4
31 - 60		3 - 4	4-5
31 - 60	Barge	2 - 3	4-5
> 60		4	≥5
> 60	Barge	2-3	≥5
> 60	Catamaran	NA	1-2